UNIVERSITY OF THE FREE STATE

BLOEMFONTEIN CAMPUS

CSIS3764

DEPARTMENT: COMPUTER SCIENCE AND INFORMATICS

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SEMESTER TEST 2
21 October 2022

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TIME: 180 min **MARKS:** 90

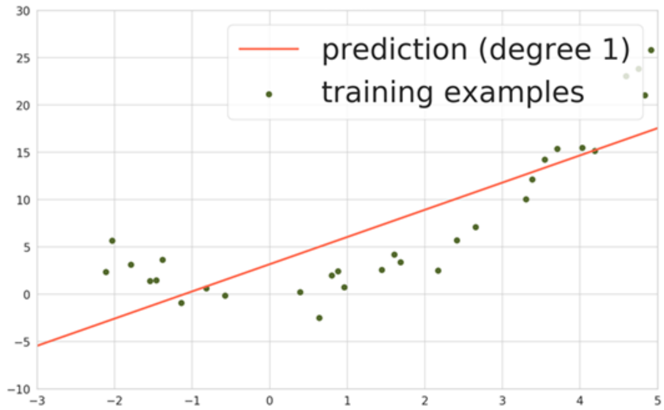
INSTRUCTIONS:

- This question paper has two questions:
  - Answer Question 1 on the answer sheet provided. (Closed Book)
  - Create and submit a Jupyter Notebook for Question 2. (Open Book)
- Submit the notebook to the following Blackboard link:
  - CSIS3764 -> Assessments -> *Semester Test 2 -> Question 2*.

Question 1 [27 Marks - 40 Minutes] (Closed Book)

1.1 Explain the main differences between supervised and unsupervised machine learning. (2)

1.2 Study the following graph of a machine learning prediction model that was trained, given the data points as depicted in the scatter plot:



1.2.1 Is the machine learning model overfitting the data, or is it underfitting the data? (1)

1.2.2 Motivate your answer in 1.2.1 by explaining why the model is either overfitting or underfitting the data. (2)

1.2.3 How can this occurrence be prevented in general? (3)

1.3 Explain step-by-step how the K-Nearest Neighbour algorithm operates to determine which category a new data point belongs to, given a training dataset that contains two or more categories. (4)

1.4 Consider the data table below:

High blood pressure (HBP)	**Younger than 60 (Age < 60)**	**Heart attack (HA)**
Yes	Yes	Yes
No	Yes	No
Yes	Yes	No
No	Yes	No
Yes	No	Yes
Yes	Yes	No
Yes	No	Yes

1.4.1 Using the Gini Impurity metric, determine the root feature of the decision tree. Show all your calculations. (11)

1.4.2 Draw a decision tree that is capable of determining whether a person will suffer a heart attack, given the person's blood pressure and age status. (4)

[27]

<u>Question 2 [63 Marks - 140 Minutes]</u> (Open Book)

Please log into Blackboard and go to CSIS3764 -> Assessments -> *Semester Test 2* -> Question 2 and download the data file named *medicine.csv*. The data contains information about patients who were successfully treated for the same illness using one of five medications. Create a Jupyter notebook that contains the Python code to construct machine learning classifiers that are able to determine which medication should be prescribed to future patients with the same illness, depending on the feature values for each specific patient. Name the notebook "*CSIS3764_YourStudentNumber_T2_2022*".

The Jupyter notebook should have the following functionality (NB!!! - Show the changes in the data after each operation):

2.1 Import the data file into a dataframe named "*meds*". (1)

2.2 Inspect the data by: (4)

- Displaying the first 20 records of the dataset.
- Generating a statistical summary of all the features.
- Getting a concise summary of the dataframe such as the index, data type, columns, non-null values and memory usage.
- Checking for any missing values.

2.3 Visualise the data: (3)
   - Use the seaborn library to create a combined scatter plot between Age and Ratio_Na_K (the ratio of sodium and potassium in the patients' blood) for each of the different medications (labels).

2.4 What information can be deduced from the scatter plot about the efficacy of the different medications with regard to the patients' age and the sodium-potassium ratio in their blood? (Print your answer in the Jupyter notebook) (4)

2.5 Determine and display the number of records relevant to each medication. (1)

2.6 Determine whether the data is balanced with regard to the number of records for each medication by: (4)
   - Creating a pie chart that shows the percentage of records for each medication.
   - Percentages displayed on the pie chart should be rounded to 2 decimals.

2.7 Perform the necessary operations to balance the data: (6)
   - Use oversampling to balance the number of records with regard to medication.
   - Assign the resampled data to the dataframe named "*meds_resampled*".
   - Use the dataframe *meds_resampled* for the remainder of Question 2.

2.8 Confirm the successful resampling of the data by: (3)
   - Displaying the number of records for each medication.
   - Recreating the pie chart that shows the percentage of records for each medication.
   - Percentages displayed on the pie chart should be rounded to 2 decimals.

2.9 Convert the text values in the dataframe to numeric values: (6)
   - Convert the columns that contain only two unique text values to binary values [0, 1].
   - Convert the columns that contain more than two unique text values to numeric values [0, 1, 2, …, *n*], with *n* depending on the number of unique text values.

2.10 Confirm that there are no more columns of type *object*. (2)

2.11 Determine the correlation between *Medication* and the other data columns. (4)
   - Using the seaborn library, create a heatmap that depicts the correlation.

2.12 What can be deduced from the correlation values with regard to their influence on the medication that will be prescribed? (Print your answer in the Jupyter notebook) (2)

2.13 Define X and y and create a training / testing dataset of 80% / 20%: (8)
   - X = Data features.
   - y = Medication.
   - Determine the dimensions of X, y, X_train and X_test.
   - Ensure that all pre-processing is complete.

2.14 Train the following classifiers and determine the best model by using k-fold cross-validation: (8)
   - K nearest neighbour (default k-value).
   - Decision tree (default values).
   - Random forest (with 5 trees).

- o Set k = 5 for the k-fold cross-validation.
- o Report the cross-validated training accuracy and F1 scores for all the classifiers.

2.15 Select the model that produced the highest F1 score to do the following: (3)

- o Make predictions using the test dataset.
- o Provide the test accuracy score, confusion matrix and classification report of the model.

2.16 Discuss the selected model's metrics and scores with regard to the following (Print your answer in the Jupyter notebook): (4)

- o Discuss the model's training accuracy in comparison to its testing accuracy.
- o Discuss the meaning of the precision, recall and f1-scores.

[63]

End of Paper